

AIFP JSON Schemas

Document: AIFP-DOC-10 · **Status:** Conforming Artifact · **Governed by:** AIFP-1 (Doc 01)

Spec dialect: JSON Schema 2020-12 · **Base \$id :** <https://schemas.aifinpay.io/v1/>

These schemas are the canonical machine-readable definitions of every AIFP protocol object. They **MUST** stay byte-consistent with the OpenAPI 3.1 components, the AIFP-1 normative spec, and the current protocol pricing model. Where documents differ, **AIFP-1 governs**.

Each schema below is independently publishable as a `.json` file under the `$id` base. A consolidated bundle is provided in `/schemas/aifp.bundle.json`.

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Object	\$id	Source of truth
Payment Challenge	<code>payment-challenge.json</code>	AIFP-1 §6
Quote	<code>quote.json</code>	AIFP-1 §16.1
Receipt Token (claims)	<code>receipt-token.json</code>	AIFP-1 §7.3
Receipt (envelope)	<code>receipt.json</code>	AIFP-1 §16.2
Merchant	<code>merchant.json</code>	AIFP-1 §11
Wallet	<code>wallet.json</code>	AIFP-1 §13
Passport	<code>passport.json</code>	AIFP-1 §24, §10.2; Doc 03 §9
Payment	<code>payment.json</code>	AIFP-1 §16.2
Settlement	<code>settlement.json</code>	AIFP-1 §19
Webhook	<code>webhook.json</code>	AIFP-1 §9.4
Error	<code>error.json</code>	AIFP-1 §17.2

Pricing Contract

Tier	Minimum amount	Intended use
standard	\$0.00001	Simple read, single record, lightweight API request.
complex	\$0.00006	Search, aggregation, multi-source queries, higher compute.

Tier	Minimum amount	Intended use
premium	\$0.00010	AI inference, GPU workloads, deep analytics, premium data.

The AiFinPay Protocol Fee is 1% of every successful transaction. The remaining 99% is settled to the merchant, excluding any applicable payment-network or settlement costs.

Shared definitions (\$defs)

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://schemas.aifinpay.io/v1/common.json",
  "title": "AIFP Common Definitions",
  "$defs": {
    "pricingTier": {
      "type": "string",
      "enum": ["standard", "complex", "premium"],
      "description": "Protocol pricing tier for the requested action."
    },
    "tierMinimumAmount": {
      "type": "string",
      "enum": ["0.00001", "0.00006", "0.00010"],
      "description": "Minimum USD amount for the selected pricing tier."
    },
    "protocolFeeRate": {
      "type": "number",
      "const": 0.01,
      "description": "AiFinPay Protocol Fee: 1% of each successful transaction."
    },
    "merchantSettlementRate": {
      "type": "number",
      "const": 0.99,
      "description": "Merchant settlement share before external network or settlement c"
    },
    "asset": { "type": "string", "enum": ["USDC", "USDT", "PYUSD"] },
    "chain": {
      "type": "string",
      "enum": ["solana", "polygon", "avalanche", "bnb", "optimism", "arbitrum", "base",
    },
    "decimalUsd": { "type": "string", "pattern": "^[0-9]+\\.?[0-9]{2,8}$", "examples": [
    "monetaryUsd": { "type": "string", "pattern": "^[0-9]+(\\.?[0-9]{1,8})?$", "examples
    "merchantId": { "type": "string", "pattern": "^mrch_[A-Za-z0-9]+$" },
    "agentId": { "type": "string", "pattern": "^agt_[A-Za-z0-9]+$" },
    "walletId": { "type": "string", "pattern": "^wlt_[A-Za-z0-9]+$" },
    "quoteId": { "type": "string", "pattern": "^qt_[A-Za-z0-9]+$" },
    "receiptId": { "type": "string", "pattern": "^rcpt_[A-Za-z0-9]+$" },
    "passportId": { "type": "string", "pattern": "^pp_[A-Za-z0-9]+$" },
    "nonce": {
      "type": "string",
      "minLength": 22,
      "maxLength": 128,

```

```

    "pattern": "^[A-Za-z0-9_-]+$",
    "description": "Single-use anti-replay value, >=128 bits entropy from a CSPRNG (A
  }
}
}

```

Payment Challenge

```

{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://schemas.aifinpay.io/v1/payment-challenge.json",
  "title": "AIFP Payment Challenge",
  "description": "Body returned with HTTP 402 (AIFP-1 §6).",
  "type": "object",
  "required": ["version", "scheme", "merchant_id", "resource", "pricing_tier", "curren
  "properties": {
    "version": { "type": "string", "const": "1.0" },
    "scheme": { "type": "string", "const": "aifp" },
    "merchant_id": { "$ref": "common.json#/$defs/merchantId" },
    "resource": { "type": "string" },
    "pricing_tier": { "$ref": "common.json#/$defs/pricingTier" },
    "estimated_amount": { "$ref": "common.json#/$defs/decimalUsd", "description": "Deci
    "currency": { "type": "string", "const": "USD" },
    "accepted_assets": { "type": "array", "items": { "$ref": "common.json#/$defs/asset"
    "accepted_chains": { "type": "array", "items": { "$ref": "common.json#/$defs/chain"
    "nonce": { "$ref": "common.json#/$defs/nonce" },
    "expires_at": { "type": "string", "format": "date-time" },
    "quote_endpoint": { "type": "string", "format": "uri", "description": "Where the ag
    "onboarding": { "type": "boolean", "default": false, "description": "true -> AIFP-4
  },
  "additionalProperties": false
}

```

Quote

```

{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://schemas.aifinpay.io/v1/quote.json",
  "title": "AIFP Quote",
  "type": "object",
  "required": ["quote_id", "merchant_id", "resource", "pricing_tier", "amount", "curren
  "properties": {
    "quote_id": { "$ref": "common.json#/$defs/quoteId" },
    "merchant_id": { "$ref": "common.json#/$defs/merchantId" },
    "resource": { "type": "string" },
    "pricing_tier": { "$ref": "common.json#/$defs/pricingTier" },

```

```

    "amount": { "$ref": "common.json#/$defs/decimalUsd" },
    "currency": { "type": "string", "const": "USD" },
    "accepted_assets": { "type": "array", "items": { "$ref": "common.json#/$defs/asset" },
    "accepted_chains": { "type": "array", "items": { "$ref": "common.json#/$defs/chain" },
    "pay_to": { "type": "object", "additionalProperties": { "type": "string" } },
    "nonce": { "$ref": "common.json#/$defs/nonce" },
    "expires_at": { "type": "string", "format": "date-time" }
  },
  "additionalProperties": false
}

```

Receipt Token (JWT claims)

```

{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://schemas.aifinpay.io/v1/receipt-token.json",
  "title": "AIFP Receipt Token Claims",
  "description": "Decoded Ed25519 JWT claim set (AIFP-1 §7.3). Header alg MUST be EdDSA",
  "type": "object",
  "required": ["iss", "sub", "aud", "resource", "amount", "nonce", "iat", "exp", "kid"],
  "properties": {
    "iss": { "type": "string", "const": "https://api.aifinpay.io" },
    "sub": { "$ref": "common.json#/$defs/agentId" },
    "aud": { "$ref": "common.json#/$defs/merchantId" },
    "resource": { "type": "string" },
    "pricing_tier": { "$ref": "common.json#/$defs/pricingTier" },
    "amount": { "$ref": "common.json#/$defs/decimalUsd" },
    "currency": { "type": "string", "const": "USD" },
    "asset": { "$ref": "common.json#/$defs/asset" },
    "chain": { "$ref": "common.json#/$defs/chain" },
    "tx_ref": { "type": "string" },
    "receipt_id": { "$ref": "common.json#/$defs/receiptId" },
    "nonce": { "$ref": "common.json#/$defs/nonce" },
    "iat": { "type": "integer" },
    "exp": { "type": "integer", "description": "Default TTL 600s after iat." },
    "kid": { "type": "string", "examples": ["aifp-2026-06"] },
    "quota": { "type": "integer", "minimum": 1, "description": "Optional multi-use claim" },
  },
  "additionalProperties": false
}

```

Receipt (API envelope)

```

{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://schemas.aifinpay.io/v1/receipt.json",

```

```

"title": "AIFP Receipt Envelope",
"type": "object",
"required": ["receipt_id", "status"],
"properties": {
  "receipt_id": { "$ref": "common.json#/$defs/receiptId" },
  "receipt": { "type": "string", "description": "Compact EdDSA JWT." },
  "status": { "type": "string", "enum": ["settled", "settling", "expired", "revoked"] },
  "tx_ref": { "type": "string" },
  "amount": { "$ref": "common.json#/$defs/decimalUsd" },
  "protocol_fee_rate": { "$ref": "common.json#/$defs/protocolFeeRate" },
  "merchant_settlement_rate": { "$ref": "common.json#/$defs/merchantSettlementRate" },
  "protocol_fee_amount": { "$ref": "common.json#/$defs/monetaryUsd" },
  "merchant_settlement_amount": { "$ref": "common.json#/$defs/monetaryUsd" },
  "expires_at": { "type": "string", "format": "date-time" },
  "poll": { "type": "string", "description": "Present on 202 async settlement." }
},
"additionalProperties": false
}

```

Merchant

```

{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://schemas.aifinpay.io/v1/merchant.json",
  "title": "AIFP Merchant",
  "type": "object",
  "required": ["merchant_id", "name", "created_at"],
  "properties": {
    "merchant_id": { "$ref": "common.json#/$defs/merchantId" },
    "name": { "type": "string" },
    "settlement_mode": { "type": "string", "enum": ["stablecoin", "fiat", "hybrid"] },
    "protocol_fee_rate": { "$ref": "common.json#/$defs/protocolFeeRate" },
    "merchant_settlement_rate": { "$ref": "common.json#/$defs/merchantSettlementRate" },
    "resources": { "type": "object", "additionalProperties": { "$ref": "common.json#/$d" },
    "created_at": { "type": "string", "format": "date-time" }
  },
  "additionalProperties": false
}

```

Wallet

```

{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://schemas.aifinpay.io/v1/wallet.json",
  "title": "AIFP Wallet",
  "type": "object",
  "required": ["wallet_id", "agent_id", "type"],

```

```

"properties": {
  "wallet_id": { "$ref": "common.json#/$defs/walletId" },
  "agent_id": { "$ref": "common.json#/$defs/agentId" },
  "type": { "type": "string", "enum": ["custodial", "non_custodial"] },
  "balances": { "type": "object", "additionalProperties": { "type": "string" } },
  "budget": {
    "type": "object",
    "properties": {
      "window": { "type": "string", "enum": ["hour", "day", "week", "month"] },
      "cap_usd": { "$ref": "common.json#/$defs/monetaryUsd" }
    }
  }
},
"additionalProperties": false
}

```

Passport

```

{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://schemas.aifinpay.io/v1/passport.json",
  "title": "AIFP Agent Passport",
  "type": "object",
  "required": ["passport_id", "agent_id", "reputation", "risk", "trust_level"],
  "properties": {
    "passport_id": { "$ref": "common.json#/$defs/passportId" },
    "agent_id": { "$ref": "common.json#/$defs/agentId" },
    "public_key": { "type": "string", "description": "Ed25519 public key (base64url).",
    "reputation": { "type": "integer", "minimum": 0, "maximum": 1000, "default": 500 },
    "risk": { "type": "integer", "minimum": 0, "maximum": 100 },
    "trust_level": { "type": "string", "enum": ["untrusted", "basic", "verified", "ente
    "created_at": { "type": "string", "format": "date-time" }
  },
  "additionalProperties": false
}

```

Payment

```

{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://schemas.aifinpay.io/v1/payment.json",
  "title": "AIFP Payment Request",
  "type": "object",
  "required": ["quote_id", "wallet_id", "asset", "chain"],
  "properties": {
    "quote_id": { "$ref": "common.json#/$defs/quoteId" },
    "wallet_id": { "$ref": "common.json#/$defs/walletId" },

```

```
    "asset": { "$ref": "common.json#/$defs/asset" },
    "chain": { "$ref": "common.json#/$defs/chain" }
  },
  "additionalProperties": false
}
```

Settlement

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://schemas.aifinpay.io/v1/settlement.json",
  "title": "AIFP Settlement",
  "type": "object",
  "required": ["receipt_id", "status", "rail"],
  "properties": {
    "receipt_id": { "$ref": "common.json#/$defs/receiptId" },
    "status": { "type": "string", "enum": ["pending", "confirmed", "failed"] },
    "rail": { "type": "string", "enum": ["stablecoin", "fiat"] },
    "chain": { "$ref": "common.json#/$defs/chain" },
    "tx_ref": { "type": "string" },
    "confirmations": { "type": "integer", "minimum": 0 }
  },
  "additionalProperties": false
}
```

Webhook

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://schemas.aifinpay.io/v1/webhook.json",
  "title": "AIFP Webhook Event",
  "type": "object",
  "required": ["id", "type", "created", "data"],
  "properties": {
    "id": { "type": "string", "examples": ["evt_2a9f"] },
    "type": {
      "type": "string",
      "enum": ["payment.succeeded", "payment.pending", "payment.failed", "receipt.issue"]
    },
    "created": { "type": "integer" },
    "data": { "type": "object" }
  },
  "additionalProperties": false
}
```



Signature: Webhook deliveries carry AIFP-Signature: `t=<ts>,v1=<hmac>` (**HMAC-SHA256** over the raw body, using a per-merchant shared secret). Verify the HMAC and the timestamp (5-minute replay window) before trusting the payload. This is a symmetric signature — merchants do NOT need to fetch JWKS to verify webhooks (AIFP-1 §9.4, §18.4).

Error

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://schemas.aifinpay.io/v1/error.json",
  "title": "AIFP Error",
  "type": "object",
  "required": ["error"],
  "properties": {
    "error": {
      "type": "object",
      "required": ["type", "code", "message"],
      "properties": {
        "type": { "type": "string" },
        "code": {
          "type": "string",
          "enum": ["AIFP-400", "AIFP-401", "AIFP-402", "AIFP-402-ONBOARDING", "AIFP-403"],
        },
        "message": { "type": "string" },
        "request_id": { "type": "string" },
        "doc": { "type": "string", "format": "uri" }
      },
      "additionalProperties": false
    }
  },
  "additionalProperties": false
}
```

Bundle

For tooling that prefers a single file, publish `schemas/aifp.bundle.json` with all the above under a top-level `$defs`, each entry retaining its `$id`. Validation tooling (Ajv 2020, `check-jsonschema`, `quicktype`) MUST resolve `$ref`s against the `https://schemas.aifinpay.io/v1/` base.

```
# Validate an example receipt envelope against the schema
npx ajv-cli validate -s receipt.json -d examples/receipt.settled.json --spec=draft2020
```